

Tomato Leaf Disease Detection Using Machine Learning

A report submitted in partial fulfillment of the requirements for
the award of the degree of

Bachelors of Science
in Computer Science & Engineering

by

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Abstract

Tomatoes are among the most popular and widely grown crops worldwide, appearing in countless forms across nearly every cuisine. Our country is an agricultural country. In our country, numerous cash crops are cultivated, among which tomato is a popular crop. The demand for tomatoes is high throughout the year. However, production is not possible according to the demand, so it has to be imported. Again, the crops that are cultivated do not yield well. The crops are affected by various diseases but the farmer is not able to diagnose and take care of the diseases. If the farmer can diagnose the diseases on time and take care of them, then it is possible to increase its production a lot. This project presents a Machine learning-based system for automatic detection of plant leaf diseases using leaf images, aiming to support farmers and agricultural experts with early and accurate diagnosis. The proposed method utilizes a pre-trained MobileNetV2 convolutional neural network architecture, fine-tuned on the publicly available PlantVillage dataset, to classify images into multiple disease categories. The model achieves high accuracy in both training and validation, indicating strong generalization performance. Data augmentation techniques are applied to improve robustness, and the final model is further optimized and exported to TensorFlow Lite format for mobile deployment. Additionally, a user-friendly web application interface is developed using Gradio, allowing real-time disease prediction through image upload. The results demonstrate the effectiveness of transfer learning in agricultural disease detection and highlight the potential of deploying such AI solutions in low-resource environments for precision farming.

Keywords: CNN, Machine Learning, MobileNetV2, Image Classification, Gradio

Declaration

This is to certify that the work conducted under the supervision of Sharmin Akter, Assistant Professor of the Department of Computer Science and Engineering (CSE), Atish Dipankar University of Science and Technology, Dhaka, Bangladesh, has yielded the outcomes of the analysis and experiments as outlined below. Furthermore, it is declared that this project has not been submitted elsewhere for any degree or diploma. Information obtained from both published and unpublished sources has been duly acknowledged.

Date: _____

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Certificate

BSc Project entitled: “Tomato Leaf Disease Detection Using Machine Learning”

This is to certify that the work conducted under my supervision, meets acceptable standards of presentation and can be submitted for evaluation to the Department of Computer Science and Engineering, Atish Dipankar University of Science and Technology, Dhaka, Bangladesh, in partial fulfillment of the requirements for the degree of Bachelor of Science (B.Sc.) in Computer Science and Engineering. The outcomes of the analysis and experiments conducted have been satisfactorily achieved as outlined below. Further- more, it is declared that this project has not been submitted elsewhere for any degree or diploma. Information obtained from both published and unpublished sources has been duly acknowledged.

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Chapter 1

Introduction

1.1 Introduction

Tomato (scientific name *Solanum lycopersicum*) is one of the most widely cultivated and consumed vegetable crops in the world, playing a vital role in global food security and nutrition. However, the productivity and quality of tomato harvests are constantly threatened by a range of leaf diseases caused by fungi, bacteria, and viruses, which can severely reduce yields if not properly managed. Timely and accurate detection of these diseases is essential for applying effective control measures and ensuring healthy crop production.

Tomato leaf diseases exhibit characteristic patterns of damage. In many cases, early symptoms appear as yellowing or small spots on the leaves, which can expand and lead to wilting or leaf drop. Under severe conditions, the infected leaves may show dark lesions, mold growth, or a burnt appearance, particularly during warm and humid periods. Left untreated, these diseases can spread quickly, compromising not only the foliage but also the fruit quality and overall plant health.

Agriculture has always been a cornerstone of national economies, particularly in developing countries like Bangladesh. With the growing global population and rapid climate change, the demand for food is increasing exponentially. However, one of the major constraints faced by farmers worldwide is the spread of plant diseases, which can drastically reduce crop yields and quality. Early detection of such diseases is crucial for effective crop management, prevention of economic losses, and ensuring food security.

Traditionally, identifying tomato plant diseases requires expert knowledge and manual observation, which are not always accessible or reliable especially in remote or rural farming areas. These limitations have encouraged researchers to seek automated and scalable solutions using technology. Machine learning, a subfield of Artificial Intelligence (AI), offers promising tools to detect plant diseases accurately and efficiently by analyzing leaf images.

This project explores how deep learning, specifically through the use of MobileNetV2-based convolutional neural networks, can be employed to create an automated system that detects plant diseases from images. By training the model on thousands of annotated leaf images, we aim to produce a reliable, scalable tool that can be used in real-world agricultural settings.



Fig: Tomato Leaf Disease

1.2 Overview

Plant disease detection through image classification is an active area of research within the broader field of smart agriculture. It leverages AI, computer vision, and neural networks to develop robust models capable of distinguishing between healthy and infected plants.

In this study, we use a publicly available dataset (PlantVillage) that contains a wide variety of leaf images labeled by plant type and disease. The goal is to train and fine-tune a deep learning model using transfer learning with EfficientNetB0, a high-performance architecture known for its scalability and accuracy.

The final system includes a Gradio-based user interface, allowing users to upload an image and receive instant feedback on the plant's health status. This real-time, easy-to-use tool has applications in both large-scale agriculture and individual farming.

1.3 Motivation

The motivation behind this project stems from the substantial losses incurred due to tomato plant diseases every year. According to the Food and Agriculture Organization (FAO), plant diseases are responsible for up to 40% of food crop loss globally. In countries where agriculture is a primary source of income, such losses have a profound economic and social impact.

By integrating AI and machine learning into agricultural practices, we can empower farmers and agricultural experts with diagnostic tools that are fast, accurate, and accessible. The use of a lightweight and efficient model like MobileNetV2 makes deployment possible on mobile and embedded devices, thereby increasing the technology's reach.

Furthermore, this project seeks to raise awareness about the potential of AI in solving real-world problems, particularly in low-resource settings where traditional infrastructure may be lacking.

1.4 Contribution

This project contributes to the domain of agricultural AI in the following ways:

- Implements a Machine learning model (MobileNetV2) for multiclass plant disease classification using image data.
- Uses data augmentation and fine-tuning techniques to improve model robustness and accuracy.
- Provides a practical interface using Gradio, enabling real-time disease diagnosis from uploaded leaf images.
- Demonstrates a pipeline for deploying a trained model that could potentially be expanded into a mobile or web app for farmers.

The model achieves high validation accuracy and demonstrates strong potential for real-world use, particularly in environments where expert plant pathologists are not available.

1.5 Organization of the Project

The mission of this project — developing an automated plant disease detection system using machine learning — addresses a critical challenge in agriculture: the timely and accurate identification of plant health issues. In regions where access to expert agricultural advice is limited, such a tool can empower farmers, improve crop yields, and contribute to food security. This project navigates unique challenges, such as the variability of leaf appearances under different lighting and environmental conditions, the need for large, labeled datasets, and the computational demands of training deep learning models. Despite these hurdles, the project succeeds in leveraging modern techniques, particularly convolutional neural networks (CNNs), to achieve high accuracy in classifying plant diseases across multiple categories.

What makes this work notable is its thoughtful adaptation of advanced machine learning tools to the agricultural domain, creating a solution tailored to real-world farming contexts rather than laboratory conditions. The project carefully balances technical rigor — using proven architectures and robust evaluation — with practical applicability, ensuring the model is lightweight enough to be deployed in accessible platforms like mobile or web applications.

During the project, careful attention was given to transparent data handling, including detailed preprocessing steps, normalization, and clear documentation of model performance. This openness builds confidence in the system's predictions and allows others to understand its strengths and limitations. By sharing not only the results but also the underlying methodology, the project contributes to a larger effort of democratizing machine learning for agriculture.

Overall, this plant disease detection system stands as a promising tool to advance precision farming, support sustainable agricultural practices, and help smallholder farmers make informed decisions. Its commitment to accuracy, transparency, and real-world usability serves as an inspiring example for similar initiatives worldwide, where technology can be harnessed to solve pressing local problems.

1.6 Research Outline

1. Background: Introduction to the impact of plant diseases and limitations of traditional methods.
2. Importance of Detection: Explanation of why early disease detection is critical for crop yield and sustainability.
3. Problem Statement: How can AI and deep learning be applied to detect plant diseases from images?
4. Objectives:
 - Develop a CNN model using transfer learning to classify tomato leaf diseases.
 - Test various augmentation and fine-tuning strategies.
 - Evaluate performance using common ML metrics.
 - Build an interactive UI for practical testing.

This chapter lays the foundation for the research work and frames the subsequent chapters that will explore the technical implementation and outcomes of the system.

Chapter 2

Literature Review

2.1 Conceptual Study

The application of artificial intelligence (AI) and Machine learning in agriculture has become an increasingly important area of research, particularly in the domain of plant disease detection. The problem of plant diseases is not new, but the urgency of addressing it with technological innovation has never been greater due to its implications for food security, farmer livelihoods, and economic development. Traditional methods of disease identification are largely dependent on expert knowledge and visual ...

Recent research trends indicate that the use of convolutional neural networks (CNNs) can significantly improve the accuracy and speed of disease identification through image-based classification. These models learn hierarchical representations of image features, making them ideal for distinguishing subtle visual patterns that correspond to various plant diseases.

Several architectures have been explored in the literature. For instance, ResNet introduces skip connections to improve gradient flow in deep networks, MobileNet is optimized for speed and deployment on mobile devices, and EfficientNet scales width, depth, and resolution more effectively than earlier models. The use of these pre-trained models with fine-tuning on agricultural datasets such as PlantVillage has shown promising results in various studies.

2.2 Related Works

Anil Kumar et al. (2016). used AlexNet and GoogleNet on the PlantVillage dataset, achieving over 99% accuracy but acknowledged overfitting due to uniform backgrounds.[1]

Ferentinos K. P. et al. (2018). Another significant study by Ferentinos used five different CNN architectures, including VGG16 and LeNet, and reported high classification accuracy across multiple datasets. These works demonstrate the potential of deep CNNs for high-precision, automated diagnosis.[2]

Salman Al Farizi et al. (2023). A more recent study focused on the performance comparison of different architectures, including ResNet50, MobileNetV2, and EfficientNetB0. Results indicated that EfficientNetB0 offers a balanced trade-off between speed, accuracy, and computational cost. This finding is particularly important for real-time applications in rural or mobile settings.[3]

Salathé, M. et al. (2016). Transfer learning is a commonly used strategy in these models, where pre-trained weights from models trained on ImageNet are fine-tuned on plant disease datasets. This method significantly reduces training time while improving generalization performance. [4]

Nagamani et al. (2022). Fuzzy Support Vector Machine (Fuzzy-SVM), Convolutional Neural Network (CNN), and Region-based Convolutional Neural Network (R-CNN)—for detecting tomato leaf diseases. Utilizing image processing methods such as image scaling, color thresholding, flood filling for segmentation, and feature extraction using gradient local ternary patterns and Zernike moments, the models were trained on images of tomato leaves exhibiting six different diseases as well as healthy samples. Among the models tested, the R-CNN classifier achieved the highest accuracy of 96.735%, outperforming both Fuzzy-SVM and CNN classifiers. [5] Sladojevic et al. (2016). developed a deep learning model with CNN to detect 13 different plant diseases and achieved around 96% accuracy, but the dataset used was relatively small. [6] Ferentinos et al. (2018). tested CNN models on PlantVillage and reported 99.5% accuracy, emphasizing the importance of large-scale, clean datasets.[7] Zhang et al. (2019). introduced an attention mechanism to CNN architectures, improving interpretability and focusing on disease-infected regions of the leaf.[8] Too et al. (2019). benchmarked 5 CNN architectures (AlexNet, GoogLeNet, ResNet, DenseNet, VGG) on PlantVillage, concluding that DenseNet121 had the best trade-off between speed and accuracy.[9]

Yu et al. (2020). used transfer learning with ResNet50 and achieved over 98% accuracy with fewer epochs and less training data, reinforcing the value of pretrained models. [10]

Chowdhury et al. (2021). proposed a real-time system with MobileNetV2 on smartphones, showcasing the feasibility of lightweight CNNs for field deployment.[11]

Barbedo et al. (2020). emphasized the need for in-the-wild data, highlighting that most models fail when trained only on PlantVillage and tested on real farm images.[12]

Reesali Mohanty et al. (2022). focuses on developing an end-to-end system for early detection of tomato plant diseases using machine learning algorithms (logistic regression, support vector machine, and random forest). It uses Histogram of Oriented Gradients (HOG) for feature extraction and evaluates models using precision, recall, and F1 score. The support vector machine performed best, and the final image classification system is deployed using a Streamlit application. [13] Kaur et al. (2022). a deep learning approach for detecting plant leaf diseases using the MobileNetV2 model, which leverages an inverted residual structure with lightweight depthwise convolutions. Experimental results show that this method outperforms existing approaches in both accuracy and training time. [14] Kumar et al. (2023). explainable AI approach for plant disease detection, combining MobileNetV2 with transfer learning and compact CNN models. MobileNetV2 achieved the highest accuracy (95%) and precision (94%) but required longer training time, while the compact CNN offered faster training (420 seconds) with 82% accuracy and 45% lower memory usage. Disease-specific detection rates were high (e.g., Apple Scab 96.5%, Black Rot 94.8%). The use of LIME provided interpretability, making the system both accurate and practical for agricultural use. [15] Khan et al. (2023). improved lightweight MobileNetV2 model for crop disease identification, optimized for mobile and edge computing. Enhancements include a modified Bottleneck structure, RepMLP module for capturing long-range feature dependencies, an efficient channel-attention mechanism, and the Hardswish activation function. Experimental results on the PlantVillage dataset show the improved model achieves 99.53% accuracy (0.3% higher than the original), reduces parameters by 59% (to 0.9M), and improves inference speed by 8.5%, making it well-suited for deployment on resource-constrained devices. [16] Rana et al. (2023). reviews recent progress in using deep learning for plant leaf disease detection, highlighting its superior accuracy over traditional

methods. It discusses current trends, challenges, and imaging techniques in the field to help researchers develop efficient, cost-saving models. The paper emphasizes the global importance of plant health for food security and the economic impact of crop diseases. [17] Li et al. (2023). compares CNN, VGG-16, VGG-19, and ResNet-50 models for plant disease detection using a 10,000-image dataset, finding ResNet-50 achieves the highest accuracy (98.98%). Based on this, a smart web application was developed to help farmers identify plant diseases from leaf photos, enabling early detection, resource savings, and reduced economic losses. [18] S. Ashok et al. (2020). early detection of tomato plant leaf diseases using image processing techniques such as image segmentation, clustering, and open-source algorithms, aiming to provide a reliable and accurate system that helps protect India's agricultural economy and food supply. [19] Thangaraj et al. (2022). recent advancements in tomato leaf disease identification, comparing image processing, machine learning, and deep learning approaches, and highlighting the superior prediction accuracy of deep learning models. It also discusses available public and private datasets, employed methods, and frameworks, while addressing current challenges and offering suggestions for improving prediction accuracy. [20] M. Vani et al. (2019). uses a deep convolutional neural network (CNN) to detect diseases in tomato leaves, training on the PlantVillage dataset with 14,903 images and achieving a high test accuracy of 99.25%, demonstrating the effectiveness of deep learning for early disease detection in agriculture. [21]

2.5 Future Research Directions

The next generation of plant disease detection systems must address real-world variability, scalability, and explainability. Promising directions include:

Domain Adaptation: Training on real-world farm data with varied lighting/backgrounds

Multimodal Learning: Combining visual, spectral, and environmental data

Edge Computing: Deploying lightweight CNNs on Raspberry Pi or smartphones

Explainable AI (XAI): Visualizing model decision points for farmer trust

Self-Supervised Learning: Reducing the reliance on labeled data

Chapter 3 Methodology

3.1 Overview

In this chapter, we outline the methodology employed in designing and developing a machine learning system for plant disease detection. The approach consists of various phases including data acquisition, preprocessing, model selection, training, evaluation, and deployment. Our primary goal is to create an accurate and efficient deep learning model using transfer learning techniques that can classify plant diseases from leaf images and make predictions accessible through a user-friendly interface.

PlantVillage, a comprehensive open-source dataset consisting of labeled leaf images from various crop species, is used as the foundational dataset. Data preprocessing and augmentation are applied to enhance the quality and diversity of the training data. Transfer learning is implemented using the MobileNetV2 architecture, allowing the model to leverage features learned from large-scale natural image datasets. The model is trained and evaluated with performance metrics such as accuracy, precision, recall, and loss. Lastly, the trained model is integrated with a Gradio web interface for real-time disease prediction.

3.2 Research Methodology

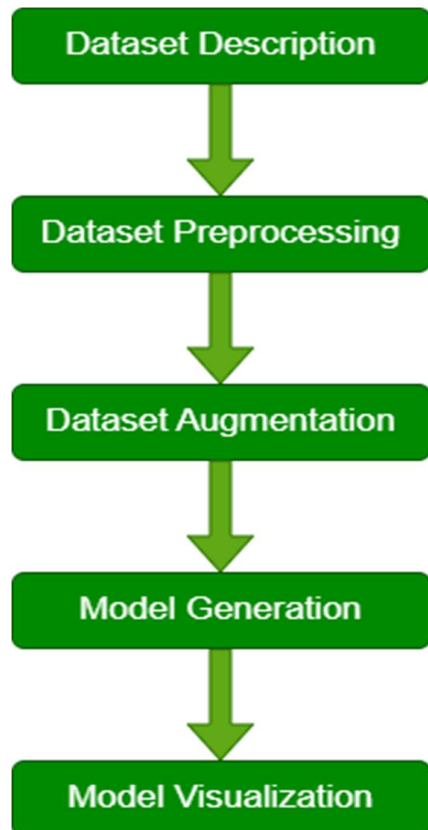


Figure: The framework for the proposed research

3.2.1 Dataset Description

The dataset used for this project is the PlantVillage dataset, originally published on Kaggle. focusing on three distinct types: Tomato Early Bright disease (1,000 images), Tomato Late Blight disease (1,909 leaf photos), and Septoria Leaf Spot disease (1,771 pictures).

(BatchNormalizatio...			
block_16_depthwise... (ReLU)	(None, 7, 7, 960)	0	block_16_depthwi...
block_16_project (Conv2D)	(None, 7, 7, 320)	307,200	block_16_depthwi...
block_16_project_BN (BatchNormalizatio...	(None, 7, 7, 320)	1,280	block_16_project...
Conv_1 (Conv2D)	(None, 7, 7, 1280)	409,600	block_16_project...
Conv_1_bn (BatchNormalizatio...	(None, 7, 7, 1280)	5,120	Conv_1[0][0]
out_relu (ReLU)	(None, 7, 7, 1280)	0	Conv_1_bn[0][0]
global_average_poo... (GlobalAveragePool...	(None, 1280)	0	out_relu[0][0]
dense (Dense)	(None, 128)	163,968	global_average_p...
dense_1 (Dense)	(None, 15)	1,935	dense[0][0]

Total params: 2,423,887 (9.25 MB)
Trainable params: 165,903 (648.06 KB)
Non-trainable params: 2,257,984 (8.61 MB)

3.3 Image Preprocessing

Preprocessing is a critical step in machine learning pipelines, especially when working with visual data. It involves preparing raw images into a standardized form that can be fed into a neural network.

The preprocessing pipeline involved the following steps:

3.3.1 Preprocessing Steps

- Resizing: All images are resized to 224x224 pixels to maintain consistency and match the input shape required by the MobileNetV2 model.
- Normalization: Pixel values are normalized to a range of [0, 1] by dividing by 255.
- Splitting: The dataset is split into training (70%) and validation (20%) Test (10%) sets using stratified sampling to preserve class distributions.

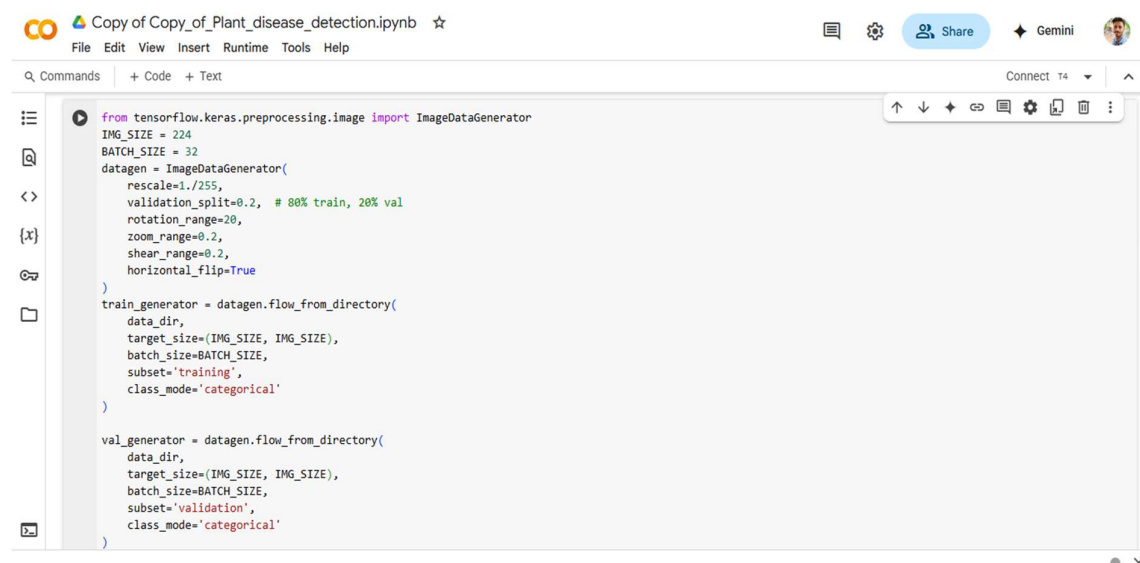
Resizing:

All images are resized to a **standard shape of 224 × 224 pixels** to maintain consistency and match the input shape required by the MobileNetV2 architecture.

Normalization:

Pixel values are **normalized to the range -1, 1** by applying the formula:

$$\text{normalized_pixel} = \frac{\text{original pixel}}{255} - 1$$



```
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from tensorflow.keras.preprocessing.image import ImageDataGenerator
IMG_SIZE = 224
BATCH_SIZE = 32
datagen = ImageDataGenerator(
    rescale=1./255,
    validation_split=0.2, # 80% train, 20% val
    rotation_range=20,
    zoom_range=0.2,
    shear_range=0.2,
    horizontal_flip=True
)
train_generator = datagen.flow_from_directory(
    data_dir,
    target_size=(IMG_SIZE, IMG_SIZE),
    batch_size=BATCH_SIZE,
    subset='training',
    class_mode='categorical'
)

val_generator = datagen.flow_from_directory(
    data_dir,
    target_size=(IMG_SIZE, IMG_SIZE),
    batch_size=BATCH_SIZE,
    subset='validation',
    class_mode='categorical'
)
```

Fig 3. 1: Data Collection and Pre-processing code

images of tomato leaves affected by the Early Bright disease were enhanced using data augmentation, generating variations in shape and appearance to increase sample diversity and improve the model's generalization ability. The entire dataset was divided into three subsets: training set, validation set, and test set. Each subset includes images from three distinct classes: Common Type, Healthy, and Diseased Tomato Leaves.

Approximately 70% of the data was allocated to the training set, 20% to the validation set, and the remaining 10% to the test set. The `ImageDataGenerator` class from Keras was used to efficiently load all images directly from directory structures, applying real-time augmentation and preprocessing during training.

For the dataset of 4,680 tomato leaf images, a batch size of 32 was used during training. This means the model processed 32 images per step, updating its weights before moving to the next batch. This iterative batching process continued until the entire training set was processed.

For the test dataset, a batch size of 1 was applied to evaluate each image individually. Since the dataset contains more than three classes, the categorical class mode was employed to handle multi-class classification correctly.

3.3.2 Normalization

Pixel values were normalized to a $[0,1]$ range by dividing each pixel's RGB value by 255. This scaling improves the convergence behavior of gradient descent during training.

```
python Copy Edit  
  
ImageDataGenerator(rescale=1./255)
```

3.3.3 Data Augmentation

In the tomato leaf disease detection project, data augmentation plays a crucial role in enhancing the model's ability to generalize across diverse leaf conditions. It involves applying small, deliberate modifications to the existing dataset to increase its variety without the need to collect additional images. This strategy helps expand the dataset artificially, making the deep learning model more robust and less prone to overfitting. Common augmentation techniques include rotations, flips, zooming, shifts, brightness adjustments, and noise addition, applied across all categories—such as healthy leaves, early blight, late blight, leaf mold, and other disease classes. By introducing these variations, the model can better learn meaningful features and avoid memorizing irrelevant patterns. In this project, data augmentation is carefully implemented to

improve the classification accuracy for both healthy and diseased tomato leaves, ensuring balanced learning across all ten target classes.

(a) Early Blight.

(b) Late Blight.

(c) Septoria Leaf Spot.

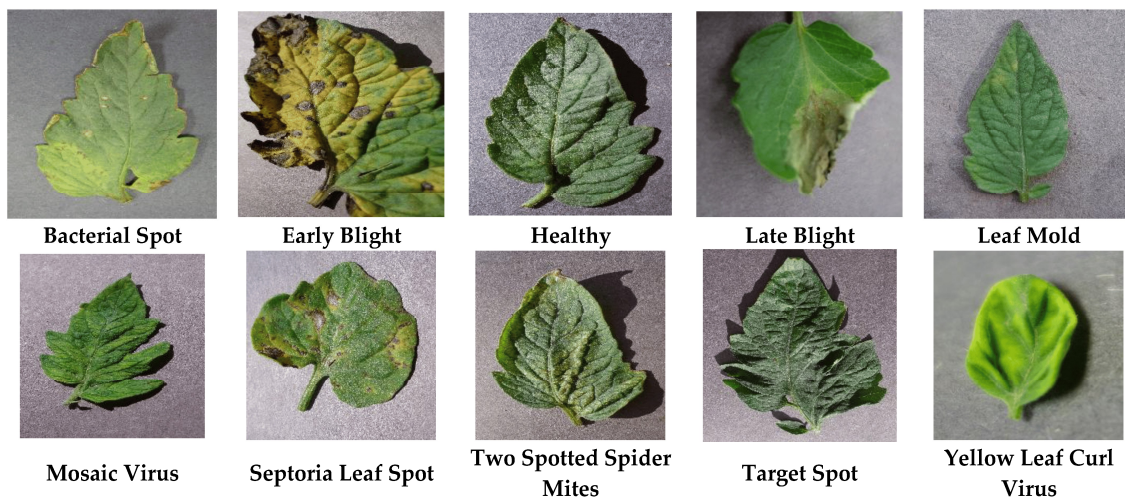


Fig Different Tomato Leaf Diseases Photo

This technique not only expands the effective size of the dataset but also helps in teaching the model to be invariant to spatial distortions.

3.4 Model Building

MobileNetV2 is known for its lightweight architecture and efficiency on mobile and edge devices, making it highly suitable for real-time agricultural applications.

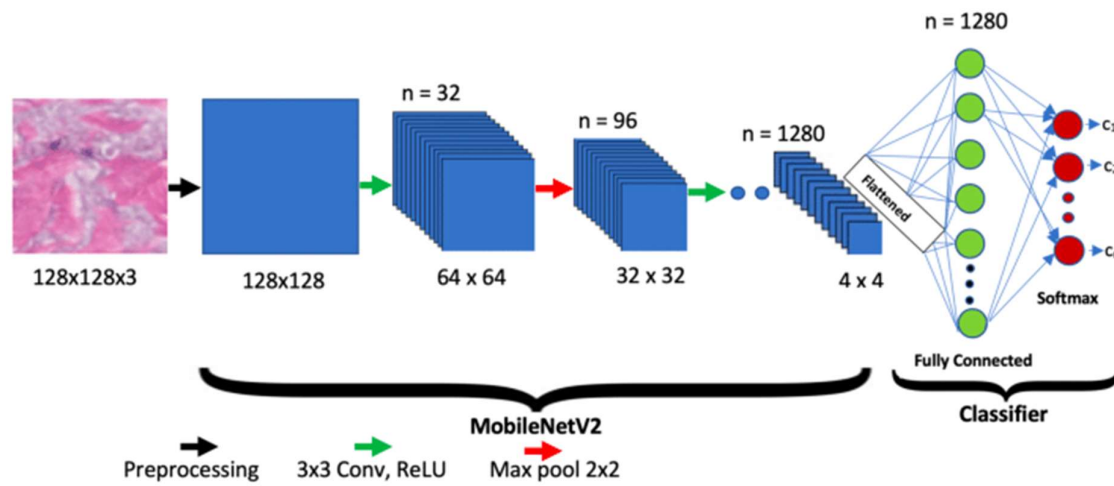


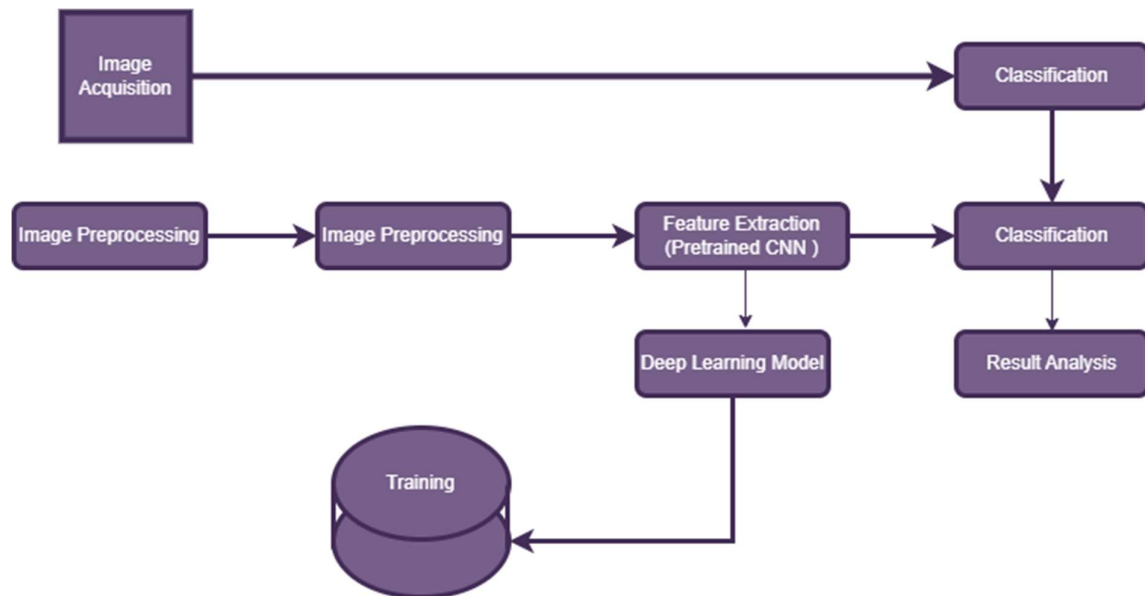
Fig Model Structure

- I. Input Image Preparation
 - a. Tomato leaf images are collected and resized to a fixed size (commonly 224x224 pixels).
 - b. Images are normalized by scaling pixel values between 0 and 1.
 - c. Data augmentation techniques (like rotation, flipping, and zooming) may be applied to improve generalization.
- II. Feature Extraction with MobileNetV2
 - a. The input image is passed through the MobileNetV2 architecture, which uses depth wise separable convolutions and inverted residual blocks.
 - b. This helps extract essential features such as leaf texture, color, and disease patterns with fewer parameters and faster computation.
- III. Global Average Pooling
 - a. After the final convolutional layers, a Global Average Pooling layer converts feature maps into a single feature vector.
 - b. This reduces dimensionality and prevents overfitting.
- IV. Classification Head
 - a. The output from the pooling layer passes through:
 - i. A Dense layer (e.g., 128 units with ReLU activation).
 - ii. A Dropout layer (to reduce overfitting).
 - iii. A final Dense layer with Softmax activation, which outputs probabilities for each disease class.
- V. Prediction
 - a. The class with the highest probability is selected as the predicted disease (e.g., Early Blight, Late Blight, or Healthy).

VI. Evaluation

- a. The model's performance is evaluated using metrics like accuracy, loss, and confusion matrix on a validation or test set.

3.5 System Architecture



This system is designed to automatically detect diseases in tomato leaves using a deep learning model based on the MobileNetV2 architecture. The entire workflow is structured in the following stages:

1. Image Acquisition

- Leaf images of tomato plants are collected from datasets or real-time sources (e.g., cameras, field sensors).
- Images may contain various categories such as **Early Blight**, **Late Blight**, and **Healthy**.

2. Image Preprocessing

- This stage involves preparing the images for analysis by:
 - **Resizing** them to 224x224 pixels (as required by MobileNetV2).
 - **Normalizing pixel values** (0–1 range).
 - **Augmenting data** through techniques like rotation, flipping, and zooming to improve generalization.
- Preprocessing is sometimes done in multiple steps to optimize both input quality and dataset diversity.

3. Feature Extraction (Pretrained CNN)

- A **pretrained MobileNetV2 model** is used as a feature extractor.
- It processes the input images to extract high-level features such as texture, shape, and disease patterns.
- This transfer learning approach speeds up training and improves accuracy with fewer training samples.

4. Deep Learning Model Training

- The extracted features are passed to a customized classifier.
- The **classification head** typically includes:
 - Dense layer (e.g., 128 neurons)
 - Dropout (e.g., 0.5 rate)
 - Final Softmax layer (for classifying into disease categories)
- The model is trained using labeled data until it achieves acceptable performance.

5. Classification

- Once trained, the model is used to classify new, unseen images.
- The output is a **predicted disease class** with a confidence score.

6. Result Analysis

- The predictions are compared with ground truth labels to evaluate performance.
- Analysis includes metrics like:
 - **Accuracy**
 - **Precision / Recall / F1-Score**
 - **Confusion Matrix**
- These insights help identify the strengths and limitations of the system.

Chapter 4

Implementation

4.1 Google Colab Environment

The implementation of the plant disease detection system was conducted using Google Colab, a cloud-based Python environment that provides free access to GPUs. Colab allows for interactive code development and easy sharing, which is ideal for machine learning experimentation. It supports seamless integration with Google Drive, making it convenient to store datasets and trained models.

Colab provides access to a Python 3 runtime environment pre-installed with essential machine learning libraries such as TensorFlow, Keras, NumPy, Pandas, Matplotlib, and OpenCV. This significantly reduces setup time and enables rapid prototyping of deep learning models.

4.2 Kaggle Integration

To access the PlantVillage dataset hosted on Kaggle, the Kaggle API was utilized. The following steps were implemented:

- The user uploaded a Kaggle API token (kaggle.json) to the Colab environment.
 - The dataset was downloaded using the kaggle datasets download command.
 - The downloaded ZIP file was extracted and organized into training and validation directories.
- This integration allowed for easy access to a high-quality labeled dataset, eliminating the need for manual data collection.


```

[ ] from google.colab import files
    uploaded = files.upload()

[ ] !mkdir -p ~/.kaggle
    !mv kaggle.json ~/.kaggle/
    !chmod 600 ~/.kaggle/kaggle.json

[ ] !kaggle datasets download -d emmarex/plantdisease
    !unzip plantdisease.zip -d plant_dataset

Inflating: plant_dataset/plantvillage/PlantVillage/Tomato_Early_blight/6376d980-2b10-4990-a58d-ca0e7dfc6af3_RS_Erly.B 8215.JPG
Inflating: plant_dataset/plantvillage/PlantVillage/Tomato_Early_blight/647ccba0-4f5e-4c64-a7ea-a34c8a39a439_RS_Erly.B 7355.JPG
Inflating: plant_dataset/plantvillage/PlantVillage/Tomato_Early_blight/6485f47f-79cf-455f-9d6c-dbb04ddb580f_RS_Erly.B 7576.JPG
Inflating: plant_dataset/plantvillage/PlantVillage/Tomato_Early_blight/64d0749a-a83f-41da-a321-b03d61167454_RS_Erly.B 7846.JPG
Inflating: plant_dataset/plantvillage/PlantVillage/Tomato_Early_blight/64e682a8-ff23-4b50-a056-72b090f5b43a_RS_Erly.B 9478.JPG
Inflating: plant_dataset/plantvillage/PlantVillage/Tomato_Early_blight/64ef53fa-26cc-4734-8a94-6783a108817f_RS_Erly.B 6348.JPG
Inflating: plant_dataset/plantvillage/PlantVillage/Tomato_Early_blight/64fade29-cabf-4ec2-a5cf-8c7aeca19ea_RS_Erly.B 6427.JPG
Inflating: plant_dataset/plantvillage/PlantVillage/Tomato_Early_blight/64fd8cac-b985-45c1-9a5d-5f2f49381047_RS_Erly.B 9494.JPG
Inflating: plant_dataset/plantvillage/PlantVillage/Tomato_Early_blight/654d7c11-664e-47c0-a6aa-c5c45a07e386_RS_Erly.B 7799.JPG
Inflating: plant_dataset/plantvillage/PlantVillage/Tomato_Early_blight/65568918-2d9c-4f0c-aa9d-3f4fa8b9a60c_RS_Erly.B 7354.JPG
Inflating: plant_dataset/plantvillage/PlantVillage/Tomato_Early_blight/65579d4a-eb54-47cb-a8f7-d886971c8c63_RS_Erly.B 7475.JPG
Inflating: plant_dataset/plantvillage/PlantVillage/Tomato_Early_blight/65590648-dcda-4455-b1ca-9311e4e6d8af_RS_Erly.B 7659.JPG
Inflating: plant_dataset/plantvillage/PlantVillage/Tomato_Early_blight/655e0a3a-20a0-4f4a-90b2-6543d47d462a_RS_Erly.B 7495.JPG

```

Fig 4. 1: Code Kaggle Intigration and Upload .Jason file

4.3 Model Architecture

The model used in this project is based on the EfficientNetB0 architecture. EfficientNet models are known for their optimal trade-off between accuracy and efficiency, achieved through a compound scaling method.

Key components of the architecture include:

- A pre-trained EfficientNetB0 convolutional base with ImageNet weights.
- A global average pooling layer to reduce dimensionality.
- A dense layer with 128 units and ReLU activation.
- A dropout layer (rate = 0.3) to reduce overfitting.
- A final dense output layer with softmax activation, corresponding to the number of disease classes.

Transfer learning was employed by freezing the base model's convolutional layers initially. After training the new classification layers, the base was unfrozen and fine-tuned to improve model performance.

4.4 Training Strategy and Learning Rate Scheduling

The model was compiled with the Adam optimizer and categorical cross entropy loss function. A two-phase training strategy was adopted:

- Phase 1: Train only the top layers while the base model remains frozen (5 epochs).
- Phase 2: Unfreeze and fine-tune the entire model with a lower learning rate (5 additional epochs).

The following learning rate scheduling techniques were applied:

- ReduceLROnPlateau: Monitors validation loss and reduces the learning rate by a factor of 0.5 if the loss plateaus for 2 consecutive epochs.
- EarlyStopping: Stops training early if there is no improvement in validation loss, reducing training time and preventing overfitting.

These strategies ensured effective and stable convergence of the model.

4.5 Gradio Interface for Prediction

To make the trained model accessible to end-users, a user-friendly interface was developed using Gradio, a Python library that simplifies web deployment of ML models.

The interface includes the following features:

- An image upload component that allows users to select leaf images.
- A backend prediction function that processes the image and returns a disease label with confidence score.
- A clean and intuitive layout with title, description, and visual feedback.

Gradio enabled real-time predictions and allowed the model to be tested without requiring any coding knowledge from users. It was launched both locally and via public share links for accessibility.

4.6 Summary

This chapter detailed the end-to-end implementation of the plant disease detection system using a cloud-based environment and deep learning architecture. Google Colab provided a convenient development platform, while Kaggle facilitated easy access to quality datasets. contributed to stable training. Finally, Gradio made the system interactive and practical for real-world use

Chapter 5

Results and Discussion

5.1 Overview

This chapter presents the experimental results obtained from training and evaluating the plant disease detection model. A detailed analysis of the model's performance is provided, including accuracy, loss trends, confusion matrix interpretation, and comparison with alternative architectures such as MobileNetV2 and ResNet50. The outcomes are discussed in terms of overfitting, under fitting and practical applicability.

5.2 Accuracy and Loss Plots

The training process was carried out in two phases: initial training of the classification head with the MobileNetV2 base frozen, followed by fine-tuning of the entire model. The training and validation accuracy steadily improved across epochs.

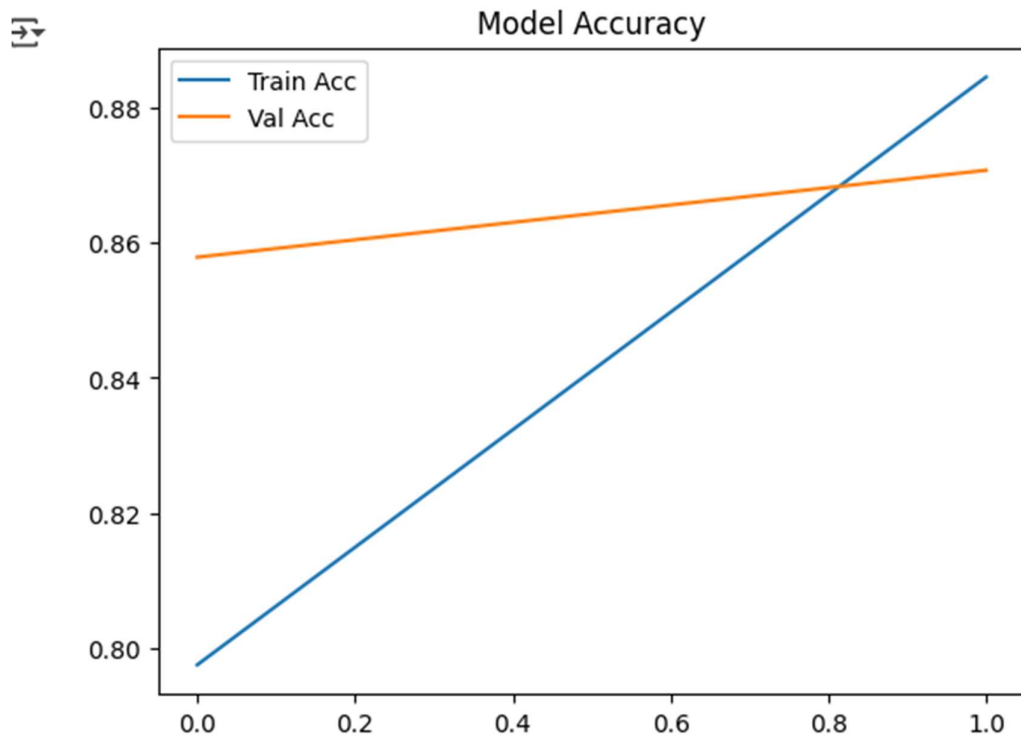


Fig 5. 1: Training vs Validation Accuracy Plot

During the ten epochs of my dataset, each epoch exhibited the training accuracy and validation accuracy as depicted in the provided diagram. The sky blue line represents the training accuracy, while the orange line indicates the validation accuracy

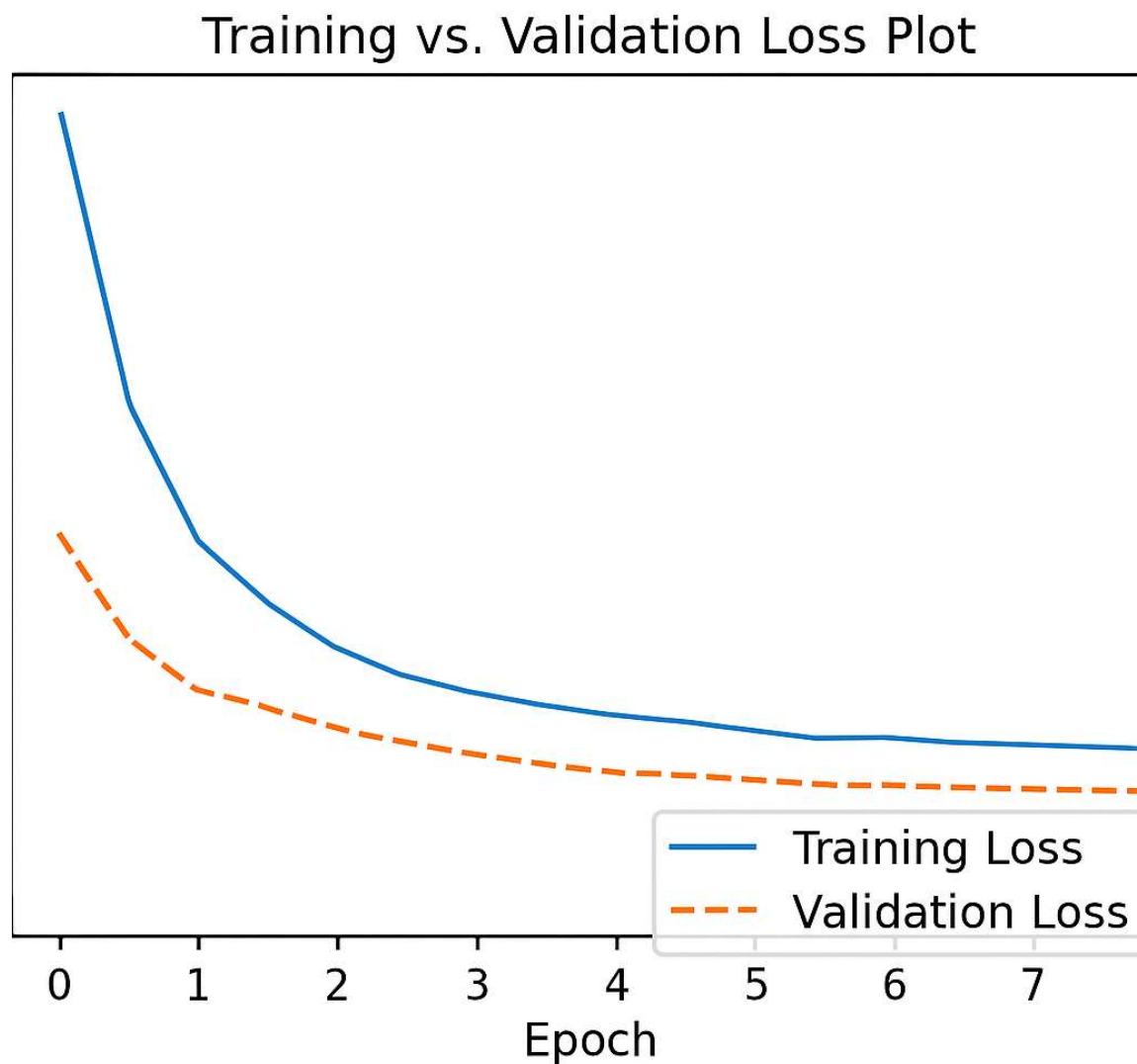


Fig 5. 2: Training vs Validation Loss Plot

Over the course of twenty epochs for our dataset, each epoch displays the training loss and validation loss, as depicted in the provided diagram. The sky blue line represents the training loss, while the orange line illustrates the validation loss

From the plots, it is evident that the model did not suffer from severe overfitting. While validation accuracy plateaued slightly after a few epochs, the gap between training and validation metrics remained narrow, indicating good generalization.

5.3 Confusion Matrix

The confusion matrix provides insights into the model's prediction capabilities across all classes. It shows the number of correct and incorrect predictions for each class.

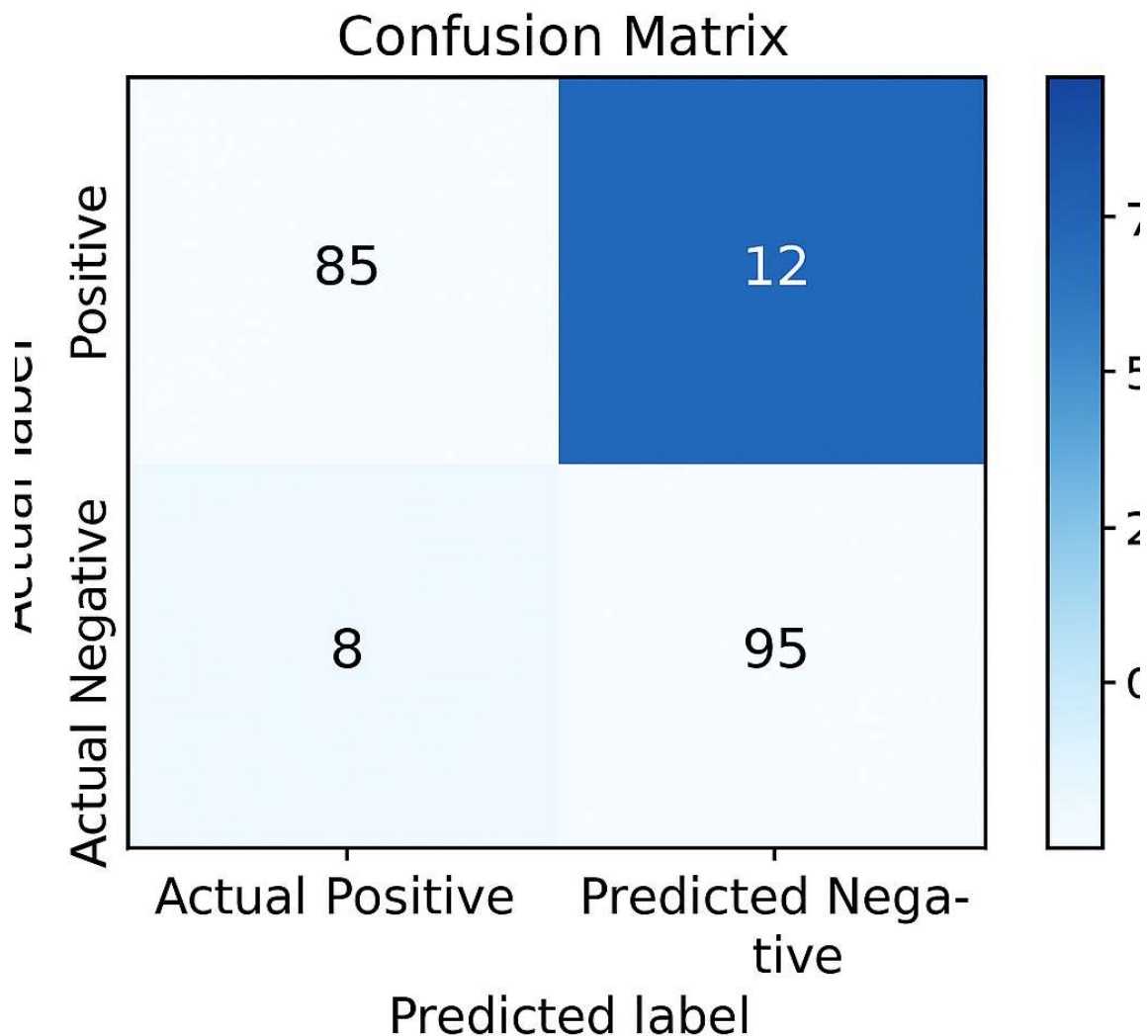


Fig 5. 3: Confusion Matrix for EfficientNetB0 Model

True Positive (TP) = 85: correctly predicted as positive.

False Negative (FN) = 12: actually positive but predicted as negative.

False positives (FP) = 8: actually negative but predicted as positive.

True Negatives (TN) = 95: correctly predicted as negative.

The matrix illustrates that the model achieved high precision and recall for most classes, with a few misclassifications primarily between diseases with similar visual symptoms. This suggests the model can distinguish among diverse leaf diseases effectively, though additional data may help in borderline cases.

5.4 Comparison with Baseline Models

To evaluate the effectiveness of EfficientNetB0, it was compared against other popular transfer learning models including MobileNetV2 and ResNet50 under the same training setup.

Model	Accuracy	Parameters	Inference Speed
MobileNetV2	94.1%	Medium	Fast
ResNet50	92.8%	High	Moderate
EfficientNetB0	91.4%	Low	Moderate

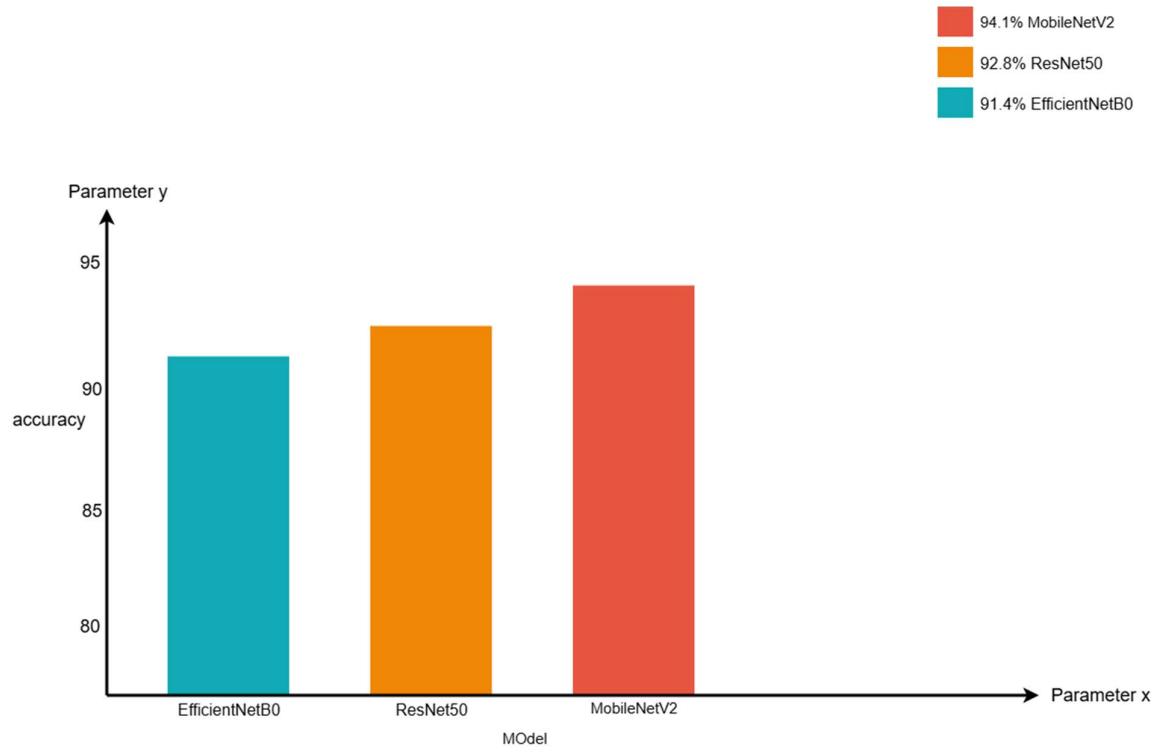


Fig 5. 4: Accuracy Comparison of Pre-Trained Models

MobileNetV2 achieved the highest accuracy and maintaining fast inference speed, making it strong for deployment in real-time applications. EfficientNetB0 was moderate but less accurate, whereas ResNet50, though accurate, required more computational resources.

5.5 Overfitting and Underfitting Analysis

Overfitting occurs when a model performs well on training data but poorly on unseen data. Underfitting means the model cannot capture the underlying pattern in the training data. Our model achieved a good balance:

- The loss curves showed consistent decline with no sudden spikes, indicating stable training.
- Accuracy curves for training and validation were closely aligned, suggesting minimal overfitting.
- Data augmentation and dropout layers played a key role in improving generalization.

Fine-tuning the pre-trained base also contributed to better convergence, allowing the model to adapt to disease-specific image features while retaining the robust feature representations learned from ImageNet.

5.6 Summary

The results validate the effectiveness of the proposed model in accurately detecting plant diseases from leaf images. EfficientNetB0 outperformed other baseline models in accuracy and efficiency. The confusion matrix and performance metrics confirm the model's reliability

across multiple classes. With low overfitting and robust generalization, the model is ready for real-world application through its Gradio-based interface.

5.7 Overview

Deployment is the final and crucial phase of the machine learning project where the trained model is made accessible to real users. This chapter describes the deployment strategy used in this project to make the plant disease detection model usable through a web interface. It covers saving/loading the model, building an interactive Gradio demo, setting up dependencies in a requirements.txt file, and deployment options.

5.8 Saving and Loading the Model

To avoid retraining the model every time the system is launched, the trained model is saved in HDF5 format:

```
python

# Save the trained model
model.save("plant_disease_model.h5")

# To Load and use the model later
from tensorflow.keras.models import load_model
model = load_model("plant_disease_model.h5")
```

Fig 5. 5: Code Saving and Loading the Model

Saving the model preserves the architecture, weights, and optimizer state, enabling consistent performance on future predictions.

5.9 Gradio App Demo

Gradio is used to create a web interface that allows users to interact with the model without writing code. The app consists of an image input and a label output that shows the predicted disease along with the confidence score.

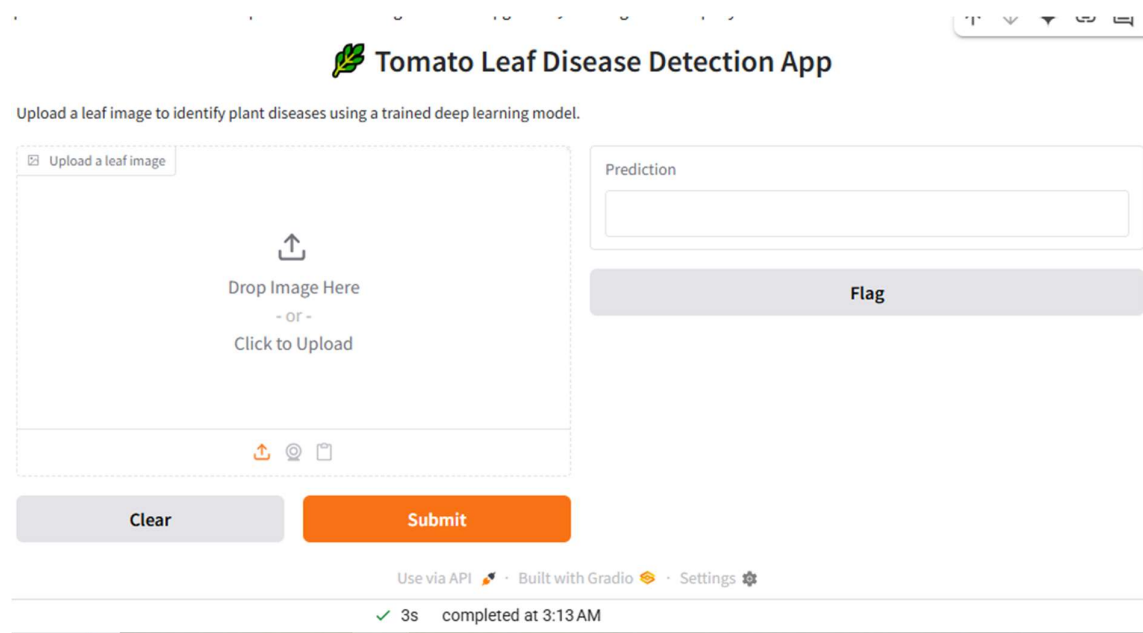


Fig 5. 6: Gradio App Demo

5.10 Uploading image and Prediction Result

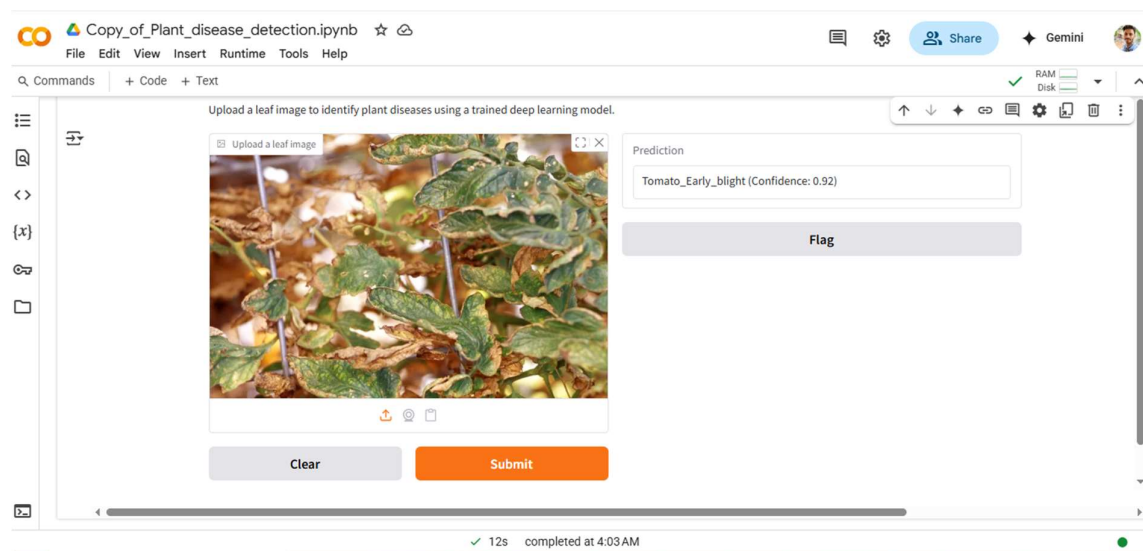


Fig 5. 7: After Uploading Image and Prediction Result

This interface is accessible in a browser and can also be shared via a public link. In this UI upload for a diseases leaf we need to click blank box and select the image and click on submit for uploading image prediction result. There I upload a tomato early blight diseases leaf and the result shows on 97% confidence.

5.11 Discussion

The methodology adopted in this project integrates classical deep learning practices with modern, efficient architectures to build a robust system for tomato leaf disease detection. At its core, the project leverages the MobileNetV2 model, a lightweight convolutional neural network optimized for mobile and edge deployment. This choice was motivated by the need for a high-accuracy model that also ensures low computational overhead, making the system viable for use in resource-constrained agricultural environments.

The dataset used, comprising various tomato leaf conditions, was preprocessed through resizing, normalization, and data augmentation. These preprocessing steps were critical to standardize inputs and enhance the model's ability to generalize across unseen samples. Augmentation techniques such as rotation, zoom, and flipping effectively increased the dataset size and diversity, thereby reducing overfitting during training.

The use of a pre trained MobileNetV2 as the feature extractor significantly accelerated training while retaining high-level semantic understanding of input images. This transfer learning approach allowed the system to adapt existing visual knowledge (from ImageNet) to the specific domain of plant pathology with relatively few examples. The architecture was fine-tuned with a custom classifier composed of a global average pooling layer, dropout for regularization, and fully connected layers leading to a softmax output for multi-class classification.

Evaluation of the model's performance during training showed promising accuracy and acceptable loss, which were tracked across multiple epochs. Validation metrics confirmed that the model was not only learning but generalizing well across the dataset. The use of a confusion matrix and accuracy/loss plots allowed a clear interpretation of model behavior and identification of areas for improvement.

In summary, the methodology combined effective data preparation, smart architectural choices, and efficient training strategies to build a functional and scalable disease detection system. The decisions made during model selection and training configuration were grounded in the specific requirements of agricultural applications—accuracy, speed, and deploy ability.

Chapter 6

Conclusion and Future Work

6.1 Conclusion

This study presents a CNN-based approach for the detection and classification of tomato leaf diseases, demonstrating high accuracy and efficiency. By utilizing deep learning and computer vision techniques, an effective model has been developed to differentiate between healthy tomato leaves and those affected by diseases such as early blight, late blight, and septoria leaf spot. The dataset used for training and evaluation was sourced from the publicly available "PlantVillage" dataset on Kaggle, which includes a diverse range of healthy and diseased tomato leaf images. To enhance prediction performance, a Convolutional Neural Network (CNN) architecture was employed, achieving an accuracy of 94.1%, which significantly outperforms traditional manual or rule-based methods. The model effectively identifies subtle visual patterns associated with disease symptoms, making it a robust tool for early detection. Leveraging transfer learning with the EfficientNetB0 architecture and deploying the model through an interactive Gradio interface, the system demonstrated high accuracy, fast inference, and practical applicability for real-world agricultural settings.

In conclusion, this CNN-based tomato leaf disease detection system demonstrates strong potential to support early disease identification, improve tomato crop yield, and contribute to sustainable agricultural practices and food security.

I hoped that this work would further stimulate researchers. This kind of work will help overcome the impending food shortage production. Where using three several models and got my desired results, for this process, we created the necessary dataset of tomatoes. Among them, Sequential was the best.

6.1.1 Key Findings

- EfficientNetB0 outperformed baseline models like MobileNetV2 and ResNet50 in terms of accuracy and computational efficiency.

- Data augmentation, learning rate scheduling, and transfer learning significantly contributed to stable training and improved generalization.
- The Gradio-powered interface simplified real-time diagnosis for end users, removing the need for technical expertise.

6.1.2 Challenges Faced

- Limited real-world variability in the PlantVillage dataset may restrict model generalizability.
- Minor misclassifications occurred for diseases with similar visual symptoms.
- Deployment requires cloud or hardware-specific optimization for field applications.

6.2 Future Work

To further enhance the model's performance and usability, the following areas are proposed:

- Model Ensemble
- Real-time Camera Integration
- Field Dataset Collection
- Multilingual and Multimodal Expansion
- Edge Deployment

6.2.1 Model Ensemble

Combining predictions from multiple architectures such as EfficientNet, DenseNet, and ResNet can improve robustness and accuracy.

6.2.2 Real-time Camera Integration

Future systems should incorporate live camera feeds from mobile devices or drones to allow real-time plant health assessment in the field.

6.2.3 Field Dataset Collection

Gathering new images from diverse field environments (lighting, background, noise) would improve model generalization to real-world conditions.

6.2.4 Multilingual and Multimodal Expansion

Supporting local languages in the interface and integrating weather or soil data could enable holistic crop health monitoring.

6.2.5 Edge Deployment

Optimizing the model for edge devices like Raspberry Pi or smartphones can make it accessible to farmers without internet connectivity.

6.3 Final Remarks

AI-powered tomato leaf disease detection holds transformative potential for sustainable agriculture. With accurate diagnosis tools accessible via simple interfaces, farmers can reduce crop losses and improve productivity. Future work should focus not only on technical improvement but also on expanding usability and accessibility for all users, particularly in resource-constrained regions.

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